Professor Q&A sheet — November 10, 2026 (Policy selection)

Phases 5A–5C, completed October 7–8, 2026

1. Why was no learned controller trained? The protocol said to train one only if a residual-complexity gate passed. Both gates failed:

- **G3B (categorical noise):** the category rule was wrong in 0.33% of validation cases.
- **G3C (continuous noise):** the best rule's material failure was 2.0% [1.6, 2.7], below the 5% threshold.

Training anyway would have been an unregistered analysis.

2. Was there no residual structure at all? There was some. It was statistically detectable (mutual information 0.08 nats in 5B, 0.04 nats in 5C validation), mostly from mean-Z readouts, but far below materiality. So: "no material residual".

3. What is a material failure? A choice whose Brier score exceeds the best by more than 20%, or that costs at least one extra shot-set per inference. Raw failure, meaning any disagreement with the Oracle, also counts near-ties and near-free overspends.

4. Doesn't the rule just read the true noise? No. Policies never see true noise parameters, labels or identifiers; a feature registry and tests enforce this. The probes are deployable measurements that estimate the noise by design. The readout probe tracks the true readout error with $\rho = 0.999$. The gate probe tracks depolarizing noise ($\rho = -0.89$) but barely tracks thermal noise ($\rho = -0.12$).

5. Is "gate noise almost always present" a finding about devices? No. It is a property of our design: 16 of 17 gate settings carry gate noise.

6. Why 64 models? To make models, nested in structures, the independent unit of analysis. That is 16 structures × 4 seeds, split 8/4/4, with the test structures sealed.

7. Do the thresholds transfer to other devices? Not without refitting. The roughly 0.4% readout threshold and the 0.97–0.98 contraction threshold are specific to this simulated 4-qubit setting.

Sources: `p5a_pool_metrics.csv`, `p5b_metadata.json`, `p5c_metadata.json`, `p5c_probe_validity.csv`, `p5c_policy_eval.csv`.